Research Synopsis

Summer Research Internship Application

Artificial Intelligence, Large Language Models, and Retrieval-Augmented Generation

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TITLE

Improving Retrieval-Augmented Generation for Accurate and Multimodal AI Systems

PROBLEM STATEMENT

Despite their strong language capabilities, Large Language Models (LLMs) often generate inaccurate or unsupported outputs — commonly referred to as hallucinations — limiting their reliability in real-world applications. This problem is most acute in knowledge-intensive domains where factual accuracy and source traceability are non-negotiable.

Retrieval-Augmented Generation (RAG) partially addresses this by grounding LLM responses in external documents. However, current RAG implementations remain constrained by imprecise retrieval, where irrelevant or redundant chunks are included in the context window, degrading answer quality. Additionally, most systems operate only on unstructured text, ignoring the rich knowledge embedded in structured sources such as relational databases and tabular records.

This research aims to design a more accurate and efficient RAG pipeline that improves retrieval precision and reduces hallucination across both structured and unstructured data sources.

PROPOSED IDEA AND RESEARCH APPROACH

The central question this work explores is: How can RAG pipelines be redesigned to generate more faithful and contextually grounded responses? To address these challenges, this work focuses on three key improvements:

1. **Adaptive Hybrid Retrieval —** Rather than relying on a single search strategy, I propose investigating ensemble retrieval that dynamically combines BM25 keyword matching with dense semantic search based on query type. A lightweight query classifier would route or blend these signals, reducing retrieval noise without sacrificing recall.
2. **Cross-Encoder Reranking and Context Pruning —** Retrieved chunks can introduce redundancy or contradiction even when individually relevant. To address this, I propose studying cross-encoder models that score query-chunk coherence jointly. This enables a reranking step that retains only the most consistent and informative context, directly targeting hallucination at its source.
3. **Structured and Multimodal Grounding —** Motivated by prior work on natural language interfaces over SQL databases, I aim to extend RAG to handle structured relational data alongside text. This involves query decomposition strategies that route sub-queries to appropriate backends (vector stores for text, SQL engines for tables), synthesizing responses from heterogeneous contexts.

MY BACKGROUND AND PREPARATION

My preparation for this internship combines hands-on system development with formal coursework in AI and machine learning. I built Piyu RAG, a production-oriented RAG system that enables natural language querying over SQL and Python documentation. The system implements hybrid retrieval (BM25 + semantic search), FlashRank reranking, and response caching, achieving a query latency of 1.24 seconds and high citation accuracy on a curated evaluation benchmark. This project gave me direct exposure to the architectural choices and failure modes of RAG pipelines — and raised the research questions this synopsis seeks to address.

Across three industry internships in data science and machine learning, I have worked on predictive modeling, feature engineering, and model deployment using PyTorch, Scikit-learn, and FastAPI. IBM certifications in Generative AI Essentials and RAG, alongside DataCamp training in prompt engineering, have strengthened my understanding of LLM-based systems. I recognize that building a working system differs from conducting rigorous research on it, and it is precisely this gap I wish to close through this internship.

TOOLS AND METHODOLOGY

The research will be implemented using modular RAG pipelines in Python, enabling controlled experimentation with retrieval, reranking, and grounding strategies. LangChain will serve as the RAG orchestration framework, with HuggingFace sentence-transformers for embedding generation and ChromaDB or FAISS as the vector store backend. PyTorch will support reranking model training where required. Evaluation will follow established RAG metrics — Recall@K, Answer Faithfulness, and Context Precision — using the RAGAS framework or custom harnesses. All experiments will be reproducible and documented to academic standards.

EXPECTED OUTCOMES

The primary expected outcome is an empirical analysis of retrieval strategy trade-offs in RAG systems, supported by a modular prototype demonstrating measurable improvement in answer faithfulness and citation accuracy over a standard vector retrieval baseline. If the structured data extension is pursued, a secondary outcome is a proof-of-concept system capable of answering queries that span both unstructured text and relational data.

This work can contribute to building more reliable AI systems for enterprise analytics, decision support, and knowledge-intensive applications. Beyond technical contributions, I aim to develop the research skills needed to formulate problems rigorously, conduct systematic experiments, and communicate findings clearly. I am committed to contributing to the lab's ongoing research agenda and to growing under the mentorship of a leading research environment. I am particularly interested in contributing to research that advances reliable and interpretable AI systems for real-world deployment.